Gabriel Choi

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EDUCATION

CALIFORNIA POLYTECHNIC STATE UNIVERSITY San Luis Obispo, CA
Bachelor of Science in **MECHANICAL ENGINEERING** **Expected May 2027**
GPA: 3.55 | Six-time President's List Award Recipient | Eagle Scout, BSA

TECHNICAL SKILLS

Design & Analysis: SolidWorks, Creo, Ansys, GD&T, DFM/DFA, MATLAB, Minitab, Excel
Testing & Validation: Test Procedure Development, X-ray Inspection, SmartScope, Instron, Rapid Prototyping
Fabrication & Programming: Additive manufacturing, CNC machining, laser cutting, waterjet cutting, Python

EXPERIENCE

Abbott Laboratories | Mechanical R&D Co-op, Cardiac Rhythm Management | Sylmar, CA **July 2026-Present**

- Apply Minitab to validate dimensional tolerances and manufacturing processes for cardiac rhythm management products.
- Design, analyze, prototype, and test components for Abbott's next generation pacemakers.
- Design single- and dual-chamber pacemaker headers for premolding manufacturing; develop and validate test procedures using X-ray inspection, SmartScope metrology, and Instron testing.

Advanced Bifurcation Systems | Mechanical/Biomedical R&D Intern | Livermore, CA **July-September 2025**

- Redesigned catheter storage system using SolidWorks and FEA analysis, transitioning from 3D-printed prototypes to laser-cut metal production units while coordinating fabrication and assembly.
- Developed resin-printed mock blood vessels and performed stent testing and analysis, modeling radial force and wall-thickness behavior with MATLAB and Excel.
- Cut manufacturing costs by 85%, reducing catheter rack expenses from **~$20,000 to <$3,000**, and replaced outsourced $1,500 mock vessels with durable $2 in-house prints.

Prototype Vehicles Laboratory | Mechanical Team Lead **Fall 2023-Present**

- Lead a 15-member mechanical team designing and manufacturing an electric vehicle for a world-record distance attempt; coordinate integration across high-voltage, low-voltage, aeroshell, and business teams.
- Designed complete battery integration system and trained 20+ members in SolidWorks, DFM, and machining processes.

PROJECTS

NASA Human Lander Challenge - Senior Design Project | Absorber System Design Lead **Fall 2025-Spring 2026**

- Selected as 1 of 12 teams and awarded $9,000 to develop MASS, an adaptive acoustic-suppression concept for NASA Artemis lander ECLSS equipment.
- Led analytical modeling and design of a tunable membrane absorber targeting 60-200 Hz noise; reduced maximum device depth from 0.305 m to 0.205 m while maintaining target-frequency coverage.
- Designed and validated a sensor-integrated actuator that swept 60-250 Hz, equalized chamber pressure in 4.49 s, and withstood simulated launch vibration to 10.3 g RMS; co-authored and presented NASA technical paper.

LEADERSHIP

Cal Poly Rotaract | Founder, Past President, Board Advisor **Spring 2024-Present**

- Founded and grew the chapter to 30+ active members; organized 40+ service events with 10+ local humanitarian organizations.
- Founded and led a Humanitarian Engineering subcommittee that developed sustainable, community-focused design solutions while expanding STEM outreach and mentorship opportunities for local K–12 students.